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Journal of Econometrics 129 (2005) 121–138
JOURNAL OF Econometrics
www.elsevier.com/locate/econbase

# Neglecting parameter changes in GARCH models

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Available online 10 February 2005

**Abstract**

If a GARCH model is estimated on a time series that contains parameter changes in the conditional volatility process and these parameter changes are not accounted for, a distinct error in the estimation occurs: The sum of the estimated autoregressive parameters of the conditional variance converges to one. In finite samples, the sum of the estimated autoregressive parameters is heavily biased towards one. This paper shows that this convergence holds for all common estimators of GARCH. Simulations of the GARCH model show that the effect occurs for realistic parameter changes and sample sizes for financial volatility data.
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*JEL classification:* C22; C51

*Keywords:* GARCH; Volatility persistence; Spurious estimation; Spurious unit root; Spurious long memory

## 1. Introduction

Estimations of GARCH models on financial returns series that span several years usually indicate high persistence: The sum of the estimated autoregressive parameters is close to one (for example, Engle and Bollerslev, 1986; Bollerslev and

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doi:10.1016/j.jeconom.2004.09.005

Engle, 1993; Baillie et al., 1996; Ding and Granger, 1996; Andersen and Bollerslev, 1997; Engle and Patton, 2001). The Integrated GARCH model was proposed to specifically reflect this fact (Engle and Bollerslev, 1986). Alternative models for financial volatility, in particular models of fractional integration, also indicate high persistence (for example, Andersen et al., 2001; Baillie et al., 1996; Ding et al., 1993; Ding and Granger, 1996). Together, these findings establish the stylized fact that financial volatility exhibits high persistence.

There is strong evidence for the occurrence of structural breaks in financial time series, as documented, for example, by Bos et al. (1999) and Andreou and Ghysels (2002). The concern that structural changes may confound persistence estimation in GARCH models was first raised by Diebold (1986) in a comment to the original IGARCH paper (Engle and Bollerslev, 1986). Lamoureux and Lastrapes (1990) explore Diebold's conjecture by allowing for changing states of the constant term of the conditional variance of a GARCH(1,1) model. Hamilton and Susmel (1994) use the Markov-switching model and improve volatility forecasts of ARCH models by incorporating regime changes. Gray (1996) extends the regime-switching approach to GARCH. These locally stationary models that segment financial data obtain substantially lower estimates of persistence. Francq et al. (2001) examine ARCH processes which are subject to Markov-switching parameters in simulations. These simulations indicate that the ARCH parameters will be estimated in the neighborhood of integration when the changes are not accounted for. Mikosch and Starica (2004) demonstrate that the Whittle estimate of the autoregressive parameter of the ARMA(1,1) representation of GARCH(1,1) will be close to unity when there are change-points that are not accounted for.

In the literature on fractionally integrated models, there is a parallel discussion of the connection between structural breaks and the estimation of the parameter of fractional integration (for example, Lobato and Savin, 1998; Granger and Hyung, 1999; Bos et al., 1999; Sakoulis and Zivot, 2000; Diebold and Inoue, 2001; Granger and Teräsvirta, 2001; Choi and Zivot, 2002; Ohannissian et al., 2003). Here, neglecting structural breaks causes an overestimation of the parameter of fractional integration and thereby of persistence.

Another closely related discussion concerns overlaying time scales in financial data. Some authors have proposed stochastic volatility models that have multiple simultaneous volatility processes with different persistence characteristics (for example, Gallant and Tauchen, 2001; Fouque et al., 2003; Chernov et al., 2003). The aggregation of several time scales in stochastic volatility models leads to apparent long memory behavior, as demonstrated by LeBaron (2001). This relates to Granger's (1980) original finding that aggregating over autoregressive models with different but low-persistence structures implies long memory behavior. These findings indicate that there may be one or more short correlation scales in financial volatility, which are not easily detectable due to the presence of other scales. Some studies have found short scales, for example Andersen and Bollerslev (1997), Gallant and Tauchen (2001), and Fouque et al. (2003). Engle and Lee (1999) suggested a two-scale GARCH model that amounts to a GARCH(2,2) specification and found strong evidence in its favor.

In this paper, I will show that parameter regime changes in GARCH models that are not accounted for in global estimations cause the sum of the estimated autoregressive parameters to converge to one. This is usually taken as evidence of high persistence, but in the case of neglected parameter changes, it is spurious and a possible confounding factor in volatility analysis. I will refer to this effect as "spurious almost-integration", for lack of a better expression. Simulations will show that this effect occurs for jump sizes and sample sizes that are realistic for financial data. Regardless of the data-generating persistence, which may be quite low, in the presence of neglected parameter changes, GARCH will indicate high persistence. Therefore, in this case, GARCH is not a suitable model to measure persistence. Prior to any GARCH estimation, a change-point detection study using, for example, the method of Kokoszka and Leipus (1999, 2000) is necessary to avoid spurious almost-integration.

This phenomenon of spurious high-persistence measurement is a consequence of the geometry of the estimation problem. It is independent of the estimation method and the statistical properties of the parameter changes. Also, the result generalizes to GARCH(p,q), showing that higher-order GARCH models like the one of Engle and Lee (1999) are not a remedy for the problem.

The rest of the paper is organized as follows. Section 2 briefly describes persistence estimation in GARCH models. Section 3 presents the proof that parameter changes that are unaccounted for push the sum of the estimated autoregressive parameters of GARCH models to unity. Section 4 explores the phenomenon in simulations with realistic sample and jump sizes. Section 5 concludes.

## 2. Persistence estimation with GARCH models

Engle (1982) and Bollerslev (1986) suggested the following estimation approach. The return $r_t$ from a stock with price $S_t$ at time $t$ is modeled as

$$r_t := \log(S_{t+1}) - \log(S_t) = \mathbb{E}(r_t|\mathscr{F}_{t-1}) + \varepsilon_t = \mu(b) + \varepsilon_t,\ t = 1, \ldots, N. \tag{1}$$

Here, $\mathscr{F}_t$ denotes the filtration modeling the information set and $\mu$ is the conditional mean function with argument $b$, for example a regression $\mu(b) = x_t'b$, where $x_t$ denotes a set of independent variables. Another example of the conditional mean is simply a constant $\mu(b) \equiv \mu$. In this paper, we are interested in the parameters of the conditional variance equation only, so we can neglect this issue.

The disturbance $\varepsilon_t$ is assumed to be normally distributed, conditional on the information available at time $t-1$:

$$\varepsilon_t|\mathscr{F}_{t-1} \sim \mathscr{N}(0, h_t), \tag{2}$$

i.e. $\varepsilon_t = \eta_t\sqrt{h_t},\ \eta_t \sim \mathscr{N}(0,1)$, where $h_t$ denotes the conditional variance. The latter is determined by the difference equation

$$h_t = \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{i=1}^{p} \beta_i h_{t-i}, \tag{3}$$

where $\sum_{i=1}^{q} \alpha_i + \sum_{i=1}^{p} \beta_i \leqslant 1$. This is the GARCH(p,q) model for the conditional variance. In the case where equality holds, we have the integrated GARCH or IGARCH model.

Consider the expected value of (3) in the GARCH(1,1) case conditional on the information set $\mathscr{F}_{t-2}$,

$$\mathbb{E}_{t-2} h_t := \mathbb{E}(h_t | \mathscr{F}_{t-2}) = \omega + \alpha \mathbb{E}_{t-2} \varepsilon_{t-1}^2 + \beta \mathbb{E}_{t-2} h_{t-1}.$$

It follows from the distribution assumption in Eq. (2) that $\mathbb{E}_{t-2} \varepsilon_{t-1}^2 = h_{t-1}$. From the definition of $h_t$, $\mathbb{E}_{t-2} h_{t-1} = h_{t-1}$, because only lagged information enters the conditional variance equation. Therefore,

$$\mathbb{E}_{t-2} h_t = \omega + (\alpha + \beta) \mathbb{E}_{t-2} h_{t-1}.$$

The conditional expected value of the variance process $h_t$ is an AR(1) process with first-order autoregressive coefficient $\alpha + \beta$. The usual stationarity conditions of autoregressive models apply. That is, in the GARCH(1,1) case the sum $\alpha + \beta$ must be less than one. In the GARCH(p,q) case, the sum $\sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j$ must be smaller than one. The closer it comes to one, the higher is the persistence of the process.

To obtain the unconditional expected variance of $\varepsilon_t$, assume that the process $h_t$ is covariance-stationary. Then, $\sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j < 1$ holds and

$$\mathbb{E} h_t = \frac{\omega}{1 - \sum_{i=1}^{q} \alpha_i - \sum_{j=1}^{p} \beta_j}. \tag{4}$$

For ease of exposition I will consider the GARCH(1,1) case only. In this case,

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1} \tag{5}$$

and

$$\mathbb{E} h_t = \omega / (1 - \lambda), \tag{6}$$

where $\lambda = \alpha + \beta$. This simplification does not come at the cost of a loss of generality. All results presented in this paper generalize to the GARCH(p,q) case.

Collect all model parameters in the vector $\theta = (b, \omega, \alpha_1, \ldots, \alpha_q, \beta_1, \ldots, \beta_p)$. The log-likelihood function is given by

$$L_N(\theta) := -\frac{1}{2N} \left[ N \log(2\pi) + \sum_{t=1}^{N} \left( \log h_t(\theta) + \frac{\varepsilon_t^2(\theta)}{h_t(\theta)} \right) \right]. \tag{7}$$

The conditional variance process $h_t$ is not observable and therefore has to be estimated at the initial parameter estimate and then updated at every iteration in the maximization. The GARCH model is not restricted to the conditionally normal case. Bollerslev (1987) suggests using the $t$-distribution and treating the number of degrees of freedom as additional parameter.

Consistency and asymptotic normality of the maximum likelihood estimator can only be proven for the GARCH(1,1) case so far. The main results can be found in Weiss (1986), Bollerslev and Wooldridge (1992), and Lumsdaine (1996). There are no closed analytical expressions for the estimators.

In practice, the likelihood is maximized by numerical optimization methods. Most software packages implement a quasi-Newton method using line search and Hessian updating algorithms. For purposes of this paper, I maximize (7) using the 'dfpmin' routine from the "Numerical Recipes" (Press et al., 2002) which implements a quasi-Newton method. The gradients are computed using analytical expressions; the Hessians are approximated by finite differencing.

Simulation studies show that the log-likelihood function does not always yield a unique maximum (Doornik and Ooms, 2000; Zumbach, 2000). In Section 4, I find a strong small sample bias in the estimates of the autoregressive parameters $\alpha$ and $\beta$. This is in accordance with the findings of Lumsdaine (1995). As the sample size increases, the parameter estimates become very accurate, though. Therefore, I use the standard estimation approach in this paper.

As an example, I estimate GARCH on daily closing data of the Dow Jones Industrial Average between December 7, 1987, and October 31, 2003. The data come from Datastream and the sample size is 4000 observations. Within the class of model orders up to GARCH(3,3), the Bayes Information Criterion favors GARCH(1,1). The estimation result for the conditional variance equation is

$$h_t = \underset{(3.388\text{e-}7)}{1.17\text{e-}6} + \underset{(.011)}{.054}\,\varepsilon_{t-1}^2 + \underset{(.010)}{.936}\,h_{t-1}.$$

The numbers in parentheses are heteroskedasticity-consistent standard errors according to Bollerslev and Wooldridge (1992).

This is a result commonly reported by the GARCH studies discussed in the introduction. If time series that cover several years are considered, $\alpha$ is usually estimated in the region between 5% and 10% and $\beta$ is usually estimated in the region above 90%. The estimated sum $\hat{\lambda} = \hat{\alpha} + \hat{\beta}$ of the autoregressive parameters, here equal to .99, is typically almost one. The $t$-statistic of $\beta$ is mostly as large as we find it here. This common result is obtained not only from various stock price data but also from exchange rate or interest rate data.

## 3. Neglecting parameter changes in GARCH estimations

In this section, I provide a proof that if a GARCH model is estimated on data that contain switches in the data-generating parameters of the conditional variance equation and these switches are not accounted for, the sum of the estimated autoregressive parameters converges to one. This phenomenon is independent of the estimation method because it stems from the geometry of the estimation problem. Also, the parameter switches need not have a specific stochastic structure; a single deterministic change-point is sufficient for the effect to occur.

**Lemma 1.** *Denote by* $\mathbb{E}_0 h_t$ *the expected value of a stationary GARCH*(1,1) *model conditional on the start value* $h_0 \in \mathbb{R}$. *Then, the relation*

$$\mathbb{E}_0 h_t = \mathbb{E} h_t + \mathrm{o}(1)_N \tag{8}$$

*holds for* $t \in \{1, \ldots, N\}$, *where* $\mathbb{E}h_t = \omega/(1-\lambda)$, $\lambda = \alpha + \beta < 1$, *and* $\mathrm{o}(1)_N \to 0$ as $N \to \infty$.

**Proof.** The expected value conditional on the initial value $h_0$ is given by

$$\mathbb{E}_0 h_t = \omega + \mathbb{E}_0(\alpha\eta_{t-1}^2 + \beta)\mathbb{E}_0 h_{t-1} = \omega + \lambda \mathbb{E}_0 h_{t-1} = \omega \frac{1-\lambda^t}{1-\lambda} + \lambda^t h_0,$$

as $\mathbb{E}_0\eta_t^2 = 1$ for all $t$ and the $\eta_t$ are independent. Thus, substituting from Eq. (6),

$$|\mathbb{E}_0 h_t - \mathbb{E}h| = \left|\omega \frac{1-\lambda^t}{1-\lambda} + \lambda^t h_0 - \frac{\omega}{1-\lambda}\right| = \lambda^t \left|h_0 - \frac{\omega}{1-\lambda}\right| = \mathrm{o}(1)_N. \qquad \square$$

**Assumption 1.** The processes $\{h_t\}$ and $\{\varepsilon_t\}$ are observable without measurement error, or at least with a measurement error that is independent of the parameter estimator $\hat{\theta}$ and that vanishes with increasing sample size.

This assumption is, of course, unrealistic. The process $h_t$ is not observable and in real estimation problems $h_t$ is estimated by $\hat{h}_t(\hat{\theta})$ and $\varepsilon_t$ by $\hat{\varepsilon}_t(\hat{\theta})$. I make the conjecture that if "spurious almost-integration" occurs when $h_t$ and $\varepsilon_t$ are observable, it will also occur when there is less perfect information about the volatility condition of the market. Section 4 provides simulation evidence to support this conjecture. For the sake of notational brevity, I assume that the measurement is error-free. The case of an error that is independent of the parameter estimates does not alter the argument.

Consider the case where there are $k-1$ points in time where the variance parameters in the data-generating $\theta$ change:

$$h_t(\theta_i) = \omega_i + \alpha_i \varepsilon_{t-1}(\theta_i)^2 + \beta_i h_{t-1}(\theta_i), \quad t = N_{i-1}+1, \ldots, N_i, \tag{9}$$

where $i = 1, \ldots, k$, and let $N_0 = 0$, and $N_k = N$. Denote the parameter vector within segments by $\theta_i$. It contains the parameters $b$ of the conditional mean equation, which do not change from segment to segment, and the segment-dependent parameters of the conditional variance $\omega_i, \alpha_i, \beta_i$.

This regime-changing structure of the data-generating process is not accounted for in the estimation. The estimated variance model is

$$h_t = \hat{\omega} + \hat{\alpha}\varepsilon_{t-1}^2 + \hat{\beta} h_{t-1}. \tag{10}$$

Let $\mathbb{E}_{(i)} h_t$ denote the expected values with respect to the initial value in segment $i = 1, \ldots, k$, so that

$$\mathbb{E}_{(i)} h_t := \mathbb{E}(h_t | \mathscr{F}_{N_{i-1}}).$$

From Lemma 1 it follows that within each segment, the $h_t$ cluster around

$$\mathbb{E}h_t(\theta_i) = \omega_i/(1-\lambda_i), \tag{11}$$

where $\lambda_i = \alpha_i + \beta_i < 1$, because $\mathbb{E}_{(i)} h_t = \mathbb{E}h_t(\theta_i) + \mathrm{o}(1)_{N_i - N_{i-1}}$.

**Lemma 2.** *Let $\mathbb{E}h_t(\theta_i)$ denote the unconditional expected value of a variance series generated by the parameter vector $\theta_i$. Let $h_t$ be generated according to* (9)*. Then,*

$$\bar{h} = \frac{1}{N}\sum_{i=1}^{k}(N_i - N_{i-1})\mathbb{E}h_t(\theta_i) + \mathrm{o}(1)_N,$$
$$\overline{\varepsilon^2} = \frac{1}{N}\sum_{i=1}^{k}(N_i - N_{i-1})\mathbb{E}h_t(\theta_i) + \mathrm{o}(1)_N, \tag{12}$$

*where it is assumed that as $N \to \infty$, the segment lengths $N_i - N_{i-1} \to \infty$.*

**Proof.** For $t \in \{N_{i-1}+1, \ldots, N_i\}$, write $h_t = \mathbb{E}_{(i)}h_t + x_t$, $x_t$ being the deviation from the expected value conditional on the initial values within segments. Then, by the law of large numbers,

$$\frac{1}{N_i - N_{i-1}}\sum_{t=N_{i-1}+1}^{N_i} x_t = \mathrm{o}(1)_{N_i - N_{i-1}}.$$

The sample mean of $h$ over the entire sample can be written as

$$\begin{aligned}
\bar{h} &= \frac{1}{N}\sum_{t=1}^{N} h_t, \\
&= \frac{1}{N}\sum_{t=1}^{N_1}\mathbb{E}_{(1)}h_t + \cdots + \frac{1}{N}\sum_{t=N_{k-1}+1}^{N}\mathbb{E}_{(k)}h_t \\
&\quad + \frac{1}{N}\sum_{t=1}^{N_1} x_t + \cdots + \frac{1}{N}\sum_{t=N_{k-1}+1}^{N} x_t, \\
&= \frac{1}{N}\sum_{t=1}^{N_1}\mathbb{E}_{(1)}h_t + \cdots + \frac{1}{N}\sum_{t=N_{k-1}+1}^{N}\mathbb{E}_{(k)}h_t + \sum_{i=1}^{k}\mathrm{o}(1)_{N_i - N_{i-1}}.
\end{aligned}$$

I assume that $N_i - N_{i-1} \to \infty$ as $N \to \infty$, so I can write the last term as $\mathrm{o}(1)_N$. Applying Lemma 1, I obtain

$$\begin{aligned}
\bar{h} &= \frac{1}{N}\sum_{t=1}^{N_1}(\mathbb{E}h_t(\theta_1) + \mathrm{o}(1)_{N_1}) + \cdots + \frac{1}{N}\sum_{t=N_{k-1}+1}^{N}(\mathbb{E}h_t(\theta_k) + \mathrm{o}(1)_{N-N_{k-1}}) + \mathrm{o}(1)_N \\
&= \frac{1}{N}\sum_{i=1}^{k}(N_i - N_{i-1})\mathbb{E}h_t(\theta_i) + \frac{1}{N}\sum_{i=1}^{k}\sum_{t=N_{i-1}+1}^{N_i}\mathrm{o}(1)_{N_i - N_{i-1}} + \mathrm{o}(1)_N.
\end{aligned}$$

The second term on the right-hand side is of the order $\mathrm{o}(1/N)$. Therefore, the last two terms can be written as $\mathrm{o}(1)_N$ and the statement for the sample mean of the $h$ is proven.

In the same manner, write $\varepsilon_t^2 = \mathbb{E}_{(i)}\varepsilon_t^2 + y_t = \mathbb{E}_{(i)}h_t + y_t$ by the distribution assumption in Eq. (2). Then,

$$\overline{\varepsilon^2} = \frac{1}{N}\sum_{t=1}^{N}\varepsilon_t^2 = \frac{1}{N}\sum_{i=1}^{k}(N_i - N_{i-1})\mathbb{E}h_t(\theta_i) + \mathrm{o}(1)_N. \qquad \square$$

**Assumption 2.** The influence of a single realization of the processes $\varepsilon_t^2$ and $h_t$ on the estimator $\hat{\theta}$ vanishes with growing sample size

$$\operatorname{cov}(\hat{\theta}, \varepsilon_t^2) = \mathrm{o}(1)_N \quad \forall t$$
$$\operatorname{cov}(\hat{\theta}, h_t) = \mathrm{o}(1)_N \quad \forall t.$$

"Spurious almost-integration" is essentially a geometric phenomenon and occurs regardless of the estimation method. The only assumption I make about the estimator $\hat{\theta}$ is that its covariance with a single observation in the $\varepsilon_t^2$ series or in the $h_t$ series vanishes with growing sample size. This assumption is not restrictive, it holds for a very general class of estimators. In essence, it assumes that some central limit theorem holds. For example, if we apply the Cauchy–Schwarz inequality to the covariance of $\hat{\theta}$ and $\varepsilon_t^2$,

$$\operatorname{cov}(\hat{\theta}, \varepsilon_t^2) = \mathbb{E}[(\hat{\theta} - \mathbb{E}\hat{\theta})(\varepsilon_t^2 - \mathbb{E}\varepsilon_t^2)] \leqslant \sqrt{\operatorname{Var}(\hat{\theta})\operatorname{Var}(\varepsilon_t^2)},$$

we see that the assumption is tantamount to a vanishing variance of the estimator as the sample size increases, given that the fourth moment of the $\varepsilon_t$ series is finite (Bollerslev, 1986).

For instance, the asymptotic distribution of the maximum likelihood estimator $\hat{\theta}$ of the GARCH(1,1) parameters $\theta = (\mu, \omega, \alpha, \beta)^{\mathrm{T}}$ is given by

$$\sqrt{N}(\hat{\theta} - \theta) \sim_{N\to\infty} \mathcal{N}(0, \Sigma),$$

where $\Sigma$ is a symmetric positive definite matrix that arises from the outer product of the likelihood score (Weiss, 1986; Bollerslev and Wooldridge, 1992; Lumsdaine, 1996). Hence, the variance of $\hat{\theta}$ vanishes with $N$ and satisfies the assumption. For higher-order GARCH models there is as yet no asymptotic distribution theory. However, it is commonly assumed that the estimators have an asymptotically normal distribution.

There is a twofold dependency in the estimation of a GARCH model, one being the dependency of the estimated volatility process $\hat{h}_t$ on the estimated parameters, the other being the dependency of the estimators on the data $h_t$ and $\varepsilon_t^2$. The former dependency is addressed by the assumption of perfect measurement, the latter by the assumption of the vanishing influence of a single realization on the estimator.

**Proposition 1.** *If there are $k$ switches in the data-generating parameters of the conditional variance equation of a GARCH(1, 1) model, as specified in Eq.* (9), *and the model is estimated on the entire series without accounting for the parameter changes, then, under Assumptions* 1 *and* 2, *the condition*

$$\mathbb{E}_{(i)}\hat{\lambda} = \mathbb{E}_{(i)}(\hat{\alpha} + \hat{\beta}) = 1$$

*must hold in every single segment $i$, up to terms that vanish with growing segment sizes $N_i - N_{i-1}$, $i = 1, \ldots, k$.*

**Proof.** Subtract the sample mean from (10) and apply Assumption 1

$$h_t - \bar{h} = \hat{\alpha}(\varepsilon_{t-1}^2 - \overline{\varepsilon^2}) + \hat{\beta}(h_{t-1} - \bar{h}). \tag{13}$$

Taking expectations of (13) conditional on the initial value within segments, we have

$$\mathbb{E}_{(i)}h_t - \mathbb{E}_{(i)}\bar{h} = \mathbb{E}_{(i)}(\hat{\alpha}\varepsilon_{t-1}^2) - \mathbb{E}_{(i)}(\hat{\alpha}\overline{\varepsilon^2}) + \mathbb{E}_{(i)}(\hat{\beta}h_{t-1}) - \mathbb{E}_{(i)}(\hat{\beta}\bar{h}).$$

Use Assumption 2 and the distribution assumption in Eq. (2):

$$\mathbb{E}_{(i)}h_t - \mathbb{E}_{(i)}\bar{h} = \mathbb{E}_{(i)}\hat{\alpha}\mathbb{E}_{(i)}h_{t-1} - \mathbb{E}_{(i)}(\hat{\alpha}\overline{\varepsilon^2}) + \mathbb{E}_{(i)}\hat{\beta}\mathbb{E}_{(i)}h_{t-1} - \mathbb{E}_{(i)}(\hat{\beta}\bar{h}) + \mathrm{o}(1)_N. \tag{14}$$

Plug in $\bar{h}$ and $\overline{\varepsilon^2}$ from Lemma 2, and use Lemma 1 to obtain

$$\begin{aligned}
&\mathbb{E}h_t(\theta_i) - \frac{1}{N}\sum_{j=1}^{k}(N_j - N_{j-1})\mathbb{E}h_t(\theta_j) + \mathrm{o}(1)_N \\
&\quad = \mathbb{E}_{(i)}\hat{\alpha}(\mathbb{E}h_t(\theta_i) + \lambda^{t-N_{i-1}-1}c_i) \\
&\qquad - \mathbb{E}_{(i)}\left[\hat{\alpha}\left(\frac{1}{N}\sum_{j=1}^{k}(N_j - N_{j-1})\mathbb{E}h_t(\theta_j) + \frac{1}{N}\sum_{j=1}^{k}\sum_{t=N_{j-1}+1}^{N_j}\lambda^{t-N_{j-1}-1}c_j\right.\right. \\
&\qquad \left.\left. + \frac{1}{N}\sum_{j=1}^{k}\sum_{t=N_{j-1}+1}^{N_j}y_t\right)\right] + \mathbb{E}_{(i)}\hat{\beta}(\mathbb{E}h_t(\theta_i) + \lambda^{t-N_{i-1}-1}c_i) \\
&\qquad - \mathbb{E}_{(i)}\left[\hat{\beta}\left(\frac{1}{N}\sum_{j=1}^{k}(N_j - N_{j-1})\mathbb{E}h_t(\theta_j) + \frac{1}{N}\sum_{j=1}^{k}\sum_{t=N_{j-1}+1}^{N_j}\lambda^{t-N_{j-1}-1}c_j\right.\right. \\
&\qquad \left.\left. + \frac{1}{N}\sum_{j=1}^{k}\sum_{t=N_{j-1}+1}^{N_j}x_t\right)\right] + \mathrm{o}(1)_N,
\end{aligned} \tag{15}$$

where the last $\mathrm{o}(1)_N$ term on the left-hand side stems from the applications of Lemma 1 to $\mathbb{E}_{(i)}h_t$ and Lemma 2 to $\mathbb{E}_{(i)}\bar{h}$. The last $\mathrm{o}(1)_N$ term on the right-hand side comes from the application of Assumption 2 in Eq. (14) above. The decompositions $h_t = \mathbb{E}_{(i)}h_t + x_t$ and $\varepsilon_t^2 = \mathbb{E}_{(i)}h_t + y_t$ were introduced in the proof of Lemma 2. The coefficient

$$c_i = h_{N_{i-1}+1} - \frac{\omega_i}{1-\lambda_i} = h_{N_{i-1}+1} - \mathbb{E}h_t(\theta_i)$$

is the distance of the initial value of segment $i$ from the unconditional mean implied by $\theta_i$.

Except for the sums of the $x_t$ and $y_t$, all the terms in parentheses in (15) are deterministic.

Assumption 2 implies that

$$\mathrm{cov}(\hat{\alpha}, \varepsilon_t^2) = \mathrm{cov}(\hat{\alpha}, \mathbb{E}_{(i)}h_t) + \mathrm{cov}(\hat{\alpha}, y_t) = \mathrm{o}(1)_N,$$

and therefore

$$\frac{1}{N}\,\mathbb{E}_{(i)}\left(\hat{\alpha}\sum_{j=1}^{k}\sum_{t=N_{j-1}+1}^{N_j} y_t\right) = \frac{1}{N}\,\mathbb{E}_{(i)}\hat{\alpha}\mathbb{E}_{(i)}\left(\sum_{j=1}^{k}\sum_{t=N_{j-1}+1}^{N_j} y_t\right) + \frac{1}{N}\operatorname{cov}\left(\hat{\alpha},\sum_{j=1}^{k}\sum_{t=N_{j-1}+1}^{N_j} y_t\right) = 2\,\frac{1}{N}\sum_{j=1}^{k}\sum_{t=N_{j-1}+1}^{N_j} \mathrm{o}(1)_N = \mathrm{o}(1)_N,$$

and analogously,

$$\frac{1}{N}\,\mathbb{E}_{(i)}\left(\hat{\beta}\sum_{j=1}^{k}\sum_{t=N_{j-1}+1}^{N_j} x_t\right) = \mathrm{o}(1)_N.$$

Plugging into (15) and arranging terms, I find

$$\mathbb{E}h_t(\theta_i) - \frac{1}{N}\sum_{j=1}^{k}(N_j - N_{j-1})\mathbb{E}h_t(\theta_j) = \mathbb{E}_{(i)}(\hat{\alpha}+\hat{\beta})\left[\mathbb{E}h_t(\theta_i) - \frac{1}{N}\sum_{j=1}^{k}(N_j - N_{j-1})\mathbb{E}h_t(\theta_j)\right] + \mathrm{o}(1)_N \tag{16}$$

That is, as $N \to \infty$ and $N_i - N_{i-1} \to \infty$, in every segment $i$ the condition

$$\mathbb{E}_{(i)}\hat{\lambda} = \mathbb{E}_{(i)}(\hat{\alpha}+\hat{\beta}) = 1$$

must hold up to terms of order $\mathrm{o}(1)_N$. Note that as long as one single change-point occurs,

$$\mathbb{E}h_t(\theta_i) - \frac{1}{N}\sum_{j=1}^{k}(N_j - N_{j-1})\mathbb{E}h_t(\theta_j) \neq 0. \qquad \square$$

This proof generalizes to the GARCH(p,q) case. This generalization can be obtained from the author upon request.

In the remainder of this section, I will provide some intuition for Proposition 1. Consider the case of a single change-point in the conditional variance parameters of a GARCH(1,1) model. In each of the two segments, the realizations of the conditional volatility process are centered approximately around the unconditional, stationary mean corresponding to the parameters of that segment.[1] If a GARCH(1,1) model is estimated globally without accounting for the segmentation,

[1] "Approximately" means up to terms that vanish with growing segment length. Strictly, the stationary measure is not defined in the case of segmented data. For this reason, the largest part of the formal argument is concerned with making this approximation precise.

the resulting estimation hyperplane (parameterized by $\hat{\omega}, \hat{\alpha}, \hat{\beta}$) must go through both segment means of $h_t$. As the mean of the $\{h_t\}$ and the mean of the $\{h_{t-1}\}$ is the same for sufficiently long segments, a line connecting two different means in the $(h_t, h_{t-1})$-subspace is close to the identity. Therefore, the estimator of $\beta$ will pick up the slope of the identity and be close to one. The remaining autoregressive parameter $\hat{\alpha}$ is chosen residually such that $\hat{\alpha} + \hat{\beta} \approx 1$. The sum will always stay slightly below one to keep the estimated process $\hat{h}_t$ from exploding.

Write down the estimated GARCH(1,1) equation with the correctly measured $h_t$ and $\varepsilon_t$:

$$h_t = \hat{\omega} + \hat{\alpha}\varepsilon_{t-1}^2 + \hat{\beta}h_{t-1}, \tag{17}$$

and subtract the sample mean

$$h_t - \bar{h} = \hat{\alpha}(\varepsilon_{t-1}^2 - \overline{\varepsilon^2}) + \hat{\beta}(h_{t-1} - \bar{h}). \tag{18}$$

One might argue that with an exact measurement of $h_t$ and $\varepsilon_t$, it makes sense to plug these values into (17) and just back out the parameters $\alpha$ and $\beta$, thereby possibly finding parameter switches. However, we are primarily interested in the case where $h_t$ is unobservable, so the trivial back-out approach does not work. Therefore, I proceed with the estimation of (17). The reader may think of a measurement error in $h_t$ and $\varepsilon_t$ that is independent of the estimators and vanishes with $N$. In that case, the parameters cannot be backed out as well.

Applying Assumptions 1 and 2 and taking expectations of (18), in the proof of Proposition 1 I obtain an expression similar to the following:

$$\mathbb{E}(h_t - \bar{h}) \approx \mathbb{E}\hat{\alpha}\mathbb{E}(\varepsilon_{t-1}^2 - \overline{\varepsilon^2}) + \mathbb{E}\hat{\beta}\mathbb{E}(h_{t-1} - \bar{h}). \tag{19}$$

According to the distribution assumption in Eq. (2),

$$\mathbb{E}\varepsilon_{t-1}^2 = \mathbb{E}(\mathbb{E}_{t-2}\varepsilon_{t-1}^2) = \mathbb{E}h_{t-1} \tag{20}$$

for $t \geqslant 2$. For essentially the same reason, $\overline{\varepsilon^2} \approx \bar{h}$, as it is shown in Lemma 2. From this, $\mathbb{E}h_t \approx \mathbb{E}h_{t-1}$, and (20) we can rewrite (19) as follows:

$$\mathbb{E}(h_t - \bar{h}) \approx \mathbb{E}(\hat{\alpha} + \hat{\beta})\mathbb{E}(h_t - \bar{h}). \tag{21}$$

If there are no parameter switches in $\theta = (\mu, \omega, \alpha, \beta)'$ and (17) is the correct specification, then $\mathbb{E}h_t = \mathbb{E}\bar{h}$. Condition (21) is trivial. However, in the case where there are switches in $\theta$, Eq. (17) is misspecified. Then, the expected value of $\bar{h}$ is a weighted average of the means of $h_t$ in the different segments, as it is shown in Lemma 2. These means are the $\mathbb{E}h_t(\theta_i) = \omega_i/(1 - \alpha_i - \beta_i)$ plus terms that vanish with growing segment length, where $\theta_i$ denotes the parameter vector in segment $i$.

That is, within each segment the difference between the expected value of $h_t$ and the expected value of the global sample mean $\bar{h}$ of $h_t$ is non-zero. Hence, condition (21) is not trivial and thus

$$\mathbb{E}(\hat{\alpha} + \hat{\beta} \,|\, \text{initial value of segment } i) \approx 1,$$

up to terms that take care of the initial values of each segment and that vanish with growing segment sizes. This is the "spurious almost-integration" effect.

## 4. Simulations

In this section, I explore the implications of Proposition 1 in simulations. The simulations illustrate that the convergence stated in the proposition does occur for sample sizes and parameter jumps that are realistic for financial data.

The simulation object is a GARCH(1,1) model that is estimated on time series of different lengths that have a single parameter change-point occurring in the middle of the sample. All three parameters of the conditional variance equation of the GARCH(1,1) model, $\omega, \alpha$, and $\beta$, are changed separately.

To study parameter changes in $\omega$, I consider the model

$$\begin{aligned} r_t &= \varepsilon_t, \\ \varepsilon_t|_{\mathcal{F}_{t-1}} &\sim \mathcal{N}(0, h_t), \\ h_t &= \omega_i + .10\varepsilon_{t-1}^2 + .70h_{t-1}. \end{aligned} \tag{22}$$

The model will be estimated on four different sample sizes, $N = 200,\ 800,\ 4000$, and 10,000. The single change-point always occurs in the middle of the sample.

I consider ten parameter switches of different sizes. I try to cover a range of volatility that is realistic for financial data. Therefore, the annualized volatility $\sigma_1$ in the first segment of the simulated data, before the change-point, is always equal to 10%. For $\sigma_2$ I then take an equidistant grid of ten points ranging from 10% to 25% annualized volatility. In the GARCH(1,1) model for daily data, the annualized volatility $\sigma$ is given by $\sqrt{250\omega/(1-\alpha-\beta)}$. Inserting the grid for $\sigma_2$ and backing out the parameter under study give the grid for the data-generating parameter value in the second segment. For each jump size and sample size, I simulate 10,000 time series and estimate GARCH(1,1) globally on each series.

There is an intuitive reason to expect that the "spurious almost-integration" effect will depend on the size of the jump: In the case of a very small size of the parameter change, the two clusters of $(h_{t-1}, h_t)$-points around $\mathbb{E}h_t(\theta_1)$ and $\mathbb{E}h_t(\theta_2)$ will be very close to each other and the support of the point clusters on the $h_{t-1}$-axis will be larger than on the $h_t$-axis. The estimation hyperplane will be influenced more by the data-generating slope in the $(h_{t-1}, h_t)$-subspace than by the slope of the identity. Therefore, the smaller the jump the closer the estimation of $\beta$ will be to the data-generating parameter value.

Table 1 shows the result for the change in $\omega$. The first row shows results of the series without a parameter change. It is apparent that there is substantial small sample bias in the estimates, consistent with the bias reported in Lumsdaine (1995). For sample sizes 200 and 800, $\beta$ is largely underestimated and $\omega$ is accordingly overestimated. As we have

$$\mathbb{E}\varepsilon_t^2 = \mathbb{E}h_t = \frac{\omega}{1-\alpha-\beta}$$

and the variance of the $\varepsilon_t$ is fixed by the sample average $\sum \hat{\varepsilon}_t^2/N$ from the data, an underestimation of $\beta$ or $\alpha$ must always be compensated by an overestimation of $\omega$

Table 1
Effects of a single change-point in the constant $\omega$ of a GARCH(1,1) model on the estimators $\hat{\omega}, \hat{\alpha}, \hat{\beta}$

| $N(N_1)$ | | | 200(100) | | | | 800(400) | | | | 4000(2000) | | | | 10,000(5000) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\sigma_2$ | $\omega_1$ | $\omega_2$ | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\lambda}$ | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\lambda}$ | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\lambda}$ | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\lambda}$ |
| .100 | 8e-6 | 8.0e-6 | 2e-5 | .11 | .47 | .58 | 1e-5 | .10 | .64 | .74 | 8e-6 | .10 | .69 | .79 | 8e-6 | .10 | .70 | .80 |
| .117 | 8e-6 | 1.1e-5 | 2e-5 | .11 | .50 | .62 | 1e-5 | .10 | .70 | .80 | 8e-6 | .10 | .74 | .84 | 7e-6 | .10 | .74 | .84 |
| .133 | 8e-6 | 1.4e-5 | 2e-5 | .12 | .59 | .71 | 6e-6 | .10 | .80 | .89 | 4e-6 | .09 | .85 | .93 | 4e-6 | .09 | .85 | .93 |
| .150 | 8e-6 | 1.8e-5 | 1e-5 | .13 | .68 | .80 | 3e-6 | .08 | .87 | .95 | 1e-6 | .06 | .92 | .98 | 1e-6 | .05 | .93 | .98 |
| .167 | 8e-6 | 2.2e-5 | 1e-5 | .13 | .74 | .87 | 2e-6 | .08 | .90 | .98 | 6e-7 | .05 | .94 | .99 | 4e-7 | .04 | .95 | .99 |
| .183 | 8e-6 | 2.7e-5 | 7e-6 | .13 | .78 | .92 | 1e-6 | .08 | .91 | .99 | 4e-7 | .05 | .95 | .99 | 3e-7 | .04 | .96 | 1 |
| .200 | 8e-6 | 3.2e-5 | 5e-6 | .14 | .81 | .95 | 1e-6 | .08 | .91 | .99 | 4e-7 | .05 | .95 | 1 | 2e-7 | .04 | .96 | 1 |
| .217 | 8e-6 | 3.8e-5 | 4e-6 | .14 | .82 | .96 | 1e-6 | .08 | .91 | .99 | 3e-7 | .05 | .95 | 1 | 2e-7 | .04 | .96 | 1 |
| .233 | 8e-6 | 4.4e-5 | 4e-6 | .15 | .82 | .97 | 1e-6 | .09 | .91 | .99 | 3e-7 | .05 | .95 | 1 | 2e-7 | .04 | .96 | 1 |
| .250 | 8e-6 | 5.0e-5 | 4e-6 | .15 | .82 | .97 | 1e-6 | .09 | .90 | .99 | 3e-7 | .05 | .94 | 1 | 2e-7 | .04 | .96 | 1 |

GARCH(1,1) model with $\mu = 0$ and $h_t = \omega_i + .10\varepsilon_{t-1}^2 + .70h_{t-1}$. The change-point always occurs in the middle of the sample. The annualized standard deviation $\sigma_1$ implied by the first segment is always .10. The entries are the means of 10,000 simulations.

and vice versa. For larger samples, the estimates of all three parameters are quite accurate.

In the case of parameter changes, the estimated sum $\hat{\lambda}$ of the autoregressive parameters converges to one as the sample size increases, as stated in Proposition 1. For jumps of sufficient size, this causes "spurious almost-integration" even in samples of rather small sizes. For example, for a jump from 10% to 18% volatility and sample size $N = 800$ representing 3 years of daily data, the sum of the estimated autoregressive parameters is .99. For the sample size $N = 4000$ representing 16 years of daily data, a single jump from 10% to 15% volatility pushes the sum of the autoregressive parameters to .98. Many of the long term GARCH studies cited in the introduction cover time spans like these. It is likely that there will be at least one change-point in such a long time span, so GARCH is very likely to pick up "spurious almost-integration".

The sum $\lambda$ of the autoregressive parameters is never estimated at exactly one. Where the tables state "1", the estimate is larger than .995. The convergence is always from the left and never attains unity. The reason lies in the estimation algorithm. The conditional volatility process has to be updated for every new parameter estimate obtained in an iteration of the maximization, so any estimate that implies a $\hat{\lambda} \geqslant 1$ would blow up the estimated conditional volatility process and the negative log-likelihood with it. So the algorithm avoids $\hat{\lambda} \geqslant 1$.

To study parameter changes in $\alpha$, I consider model (22) with the conditional variance equation

$$h_t = 1.6\text{e-}6 + \alpha_i \varepsilon_{t-1}^2 + .50h_{t-1}. \tag{23}$$

In the first segment, $\alpha_1$ is always equal to .10, yielding 10% annualized volatility. The jump size grid for $\alpha_2$ is obtained as before. Table 2 shows the simulation results.

Table 2
Effects of a single change-point in the $\alpha$ parameter of a GARCH(1,1) model on the estimators $\hat{\omega}, \hat{\alpha}, \hat{\beta}$

| $N(N_1)$ | | | | 200(100) | | | | 800(400) | | | | 4000(2000) | | | | 10,000(5000) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\sigma_2$ | $\alpha_1$ | $\alpha_2$ | $\bar{\lambda}$ | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\lambda}$ | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\lambda}$ | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\lambda}$ | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\lambda}$ |
| .100 | .10 | .10 | .60 | 2e-5 | .11 | .36 | .46 | 2e-5 | .10 | .43 | .54 | 2e-5 | .10 | .49 | .59 | 2e-5 | .10 | .49 | .59 |
| .117 | .10 | .21 | .66 | 2e-5 | .16 | .41 | .57 | 2e-5 | .16 | .50 | .66 | 1e-5 | .16 | .53 | .69 | 1e-5 | .16 | .53 | .70 |
| .133 | .10 | .28 | .69 | 2e-5 | .21 | .45 | .66 | 1e-5 | .21 | .54 | .75 | 1e-5 | .21 | .56 | .77 | 1e-5 | .21 | .56 | .77 |
| .150 | .10 | .32 | .71 | 2e-5 | .24 | .48 | .72 | 1e-5 | .24 | .56 | .80 | 1e-5 | .25 | .57 | .82 | 1e-5 | .25 | .57 | .82 |
| .167 | .10 | .36 | .73 | 2e-5 | .26 | .51 | .76 | 1e-5 | .27 | .56 | .83 | 1e-5 | .28 | .57 | .85 | 1e-5 | .28 | .58 | .85 |
| .183 | .10 | .38 | .74 | 1e-5 | .28 | .51 | .79 | 1e-5 | .29 | .57 | .86 | 1e-5 | .30 | .58 | .88 | 1e-5 | .30 | .58 | .88 |
| .200 | .10 | .40 | .75 | 1e-5 | .29 | .52 | .81 | 1e-5 | .30 | .57 | .87 | 1e-5 | .31 | .58 | .89 | 1e-5 | .32 | .58 | .90 |
| .217 | .10 | .42 | .76 | 1e-5 | .30 | .52 | .83 | 1e-5 | .31 | .57 | .89 | 1e-5 | .32 | .58 | .91 | 1e-5 | .33 | .58 | .91 |
| .233 | .10 | .43 | .77 | 1e-5 | .31 | .53 | .84 | 1e-5 | .32 | .57 | .90 | 1e-5 | .33 | .58 | .92 | 1e-5 | .34 | .58 | .92 |
| .250 | .10 | .44 | .77 | 1e-5 | .32 | .53 | .85 | 1e-5 | .33 | .57 | .90 | 1e-5 | .34 | .58 | .92 | 1e-5 | .34 | .58 | .93 |

GARCH(1,1) model with $\mu = 0$ and $h_t = 1.6e-5 + \alpha_i \varepsilon_{t-1}^2 + .50 h_{t-1}$. The change-point always occurs in the middle of the sample. The annualized standard deviation $\sigma_1$ implied by the first segment is always .10. In the first segment, the parameter is always $\alpha_1 = .10$ so that the sum $\lambda_1$ of the autoregressive parameters in the first segment is always .60. The column titled $\bar{\lambda}$ reports the average $.5\lambda_1 + .5\lambda_2$. The entries are the means of 10,000 simulations.

As before, the estimated sum $\hat{\lambda}$ of the autoregressive parameters moves towards unity with growing sample size and with growing jump size. It is estimated above the mean of the in-segment data-generating $\lambda$'s and grows monotonically with sample and jump size. However, the speed of the convergence is much slower than in the case of jumps in the constant. For the sample and jump sizes considered here, $\hat{\lambda}$ does not come close to unity.

From the distribution assumption in Eq. (2) we have that

$$h_t = \omega + \alpha \eta_{t-1}^2 h_{t-1} + \beta h_{t-1},$$

where $\eta_t$ is a standard normal random variable. From Section 3 we know that the position of the point clusters in the $(h_t, h_{t-1})$-subspace is the cause of "spurious almost-integration". From the conditional variance equation above we see that the relation between $h_t$ and $h_{t-1}$ is much more direct for the term with the $\beta$ coefficient than for the term with the $\alpha$ coefficient. The $\alpha$ coefficient is cushioned by the random process, which may be one reason why changes in $\alpha$ have a less dramatic impact on $\hat{\lambda}$.

To study parameter changes in $\beta$, I consider model (22) with the conditional variance equation

$$h_t = 1.6e\text{-}5 + .10 \varepsilon_{t-1}^2 + \beta_i h_{t-1}. \tag{24}$$

As before, the range of considered volatilities in the second segment is 10–25% and the $\beta_2$ values for the second segment are chosen accordingly. Table 3 shows that changes in $\beta$ cause "spurious almost-integration" very quickly. For example, for the sample size $N = 4000$ and a switch from 10% to 13% volatility the mean of the data-generating $\lambda$'s within segments is .69. However, the switch moves $\hat{\lambda}$ to .93, whereas a switch to 15% causes "spurious almost-integration".

Table 3
Effects of a single change-point in the $\beta$ parameter of a GARCH(1,1) model on the estimators $\hat{\omega}, \hat{\alpha}, \hat{\beta}$

| $N(N_1)$ | | | | 200(100) | | | | 800(400) | | | | 4000(2000) | | | | 10,000(5000) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\sigma_2$ | $\beta_1$ | $\beta_2$ | $\bar{\lambda}$ | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\lambda}$ | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\lambda}$ | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\lambda}$ | $\hat{\omega}$ | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\lambda}$ |
| .100 | .50 | .50 | .60 | 2e-5 | .11 | .36 | .46 | 2e-5 | .10 | .43 | .54 | 2e-5 | .10 | .49 | .59 | 2e-5 | .10 | .49 | .59 |
| .117 | .50 | .61 | .66 | 2e-5 | .11 | .43 | .54 | 1e-5 | .10 | .58 | .68 | 1e-5 | .10 | .64 | .74 | 1e-5 | .10 | .64 | .74 |
| .133 | .50 | .68 | .69 | 2e-5 | .12 | .55 | .67 | 8e-6 | .09 | .76 | .86 | 4e-6 | .07 | .85 | .93 | 3e-7 | .07 | .87 | .94 |
| .150 | .50 | .72 | .71 | 1e-5 | .12 | .65 | .78 | 3e-6 | .08 | .87 | .95 | 1e-6 | .05 | .94 | .98 | 6e-7 | .04 | .96 | .99 |
| .167 | .50 | .76 | .73 | 1e-5 | .13 | .73 | .86 | 2e-6 | .08 | .90 | .98 | 6e-7 | .05 | .95 | .99 | 4e-7 | .04 | .96 | .99 |
| .183 | .50 | .78 | .74 | 7e-6 | .13 | .78 | .91 | 1e-6 | .08 | .91 | .98 | 5e-7 | .05 | .94 | .99 | 4e-7 | .04 | .95 | 1 |
| .200 | .50 | .80 | .75 | 6e-6 | .14 | .80 | .94 | 1e-6 | .08 | .91 | .99 | 5e-7 | .06 | .94 | .99 | 4e-7 | .05 | .95 | 1 |
| .217 | .50 | .81 | .76 | 5e-6 | .14 | .81 | .95 | 1e-6 | .09 | .90 | .99 | 5e-7 | .06 | .94 | 1 | 4e-7 | .05 | .94 | 1 |
| .233 | .50 | .83 | .77 | 4e-6 | .14 | .82 | .97 | 1e-6 | .09 | .90 | .99 | 5e-7 | .06 | .93 | 1 | 4e-7 | .06 | .94 | 1 |
| .250 | .50 | .84 | .77 | 4e-6 | .15 | .82 | .97 | 1e-6 | .10 | .90 | .99 | 5e-7 | .07 | .93 | 1 | 4e-7 | .06 | .94 | 1 |

GARCH(1,1) model with $\mu = 0$ and $h_t = 1.6\text{e}-5 + .10\varepsilon_{t-1}^2 + \beta_i h_{t-1}$. The change-point always occurs in the middle of the series. The annualized standard deviation $\sigma_1$ implied by the first segment is always .10. In the first segment, the parameter is always $\beta_1 = .50$ so that the sum $\lambda_1$ of the autoregressive parameters in the first segment is always .60. The column titled $\bar{\lambda}$ reports the average $.5\lambda_1 + .5\lambda_2$. The entries are the means of 10,000 simulations.

I ran the same simulations for changes in the constant $\omega$ in the data-generating process $h_t = \omega_i + .30\varepsilon_{t-1}^2 + .50h_{t-1}$. For changes in $\alpha$ and $\beta$, I started at the process $h_t = 1.6\text{e-}5 + .30\varepsilon_{t-1}^2 + .30h_{t-1}$, which implies 10% volatility, and increased the autoregressive parameters in ten steps until the volatility reached 20%. The results were very similar to the ones reported here.

Also, I simulated changes in all parameters of two GARCH(2,2) models, following the same jump size grid for the annualized volatility. The "spurious almost-integration" results were very similar to the GARCH(1,1) case. These simulations are available from the author upon request. This demonstrates that in the presence of parameter changes that are not accounted for, higher-order GARCH models are not suited to capturing multiple short time scales.

In summary, "spurious almost-integration" occurs for sample sizes and jump sizes of realistic magnitudes. Even if the estimated sum of the autoregressive parameters is not pushed close to one, it is severely overestimated if there are neglected change-points in the data. In the cases of parameter changes in the constant $\omega$ and in the second autoregressive parameter $\beta$ of the conditional variance equation, the speed of the convergence of the estimated sum of the autoregressive parameters to one is faster than in the case of changes in the first autoregressive parameter $\alpha$.

## 5. Conclusion

Neglecting changes in the data-generating parameters of the conditional variance of GARCH models causes a substantial overestimation of the autoregressive parameters of the conditional variance. In this paper, I show that this effect stems

from the geometry of the estimation problem. Different parameter regimes imply different means of volatility. In the space spanned by the time series $h_t$ and its own lags, the different means of the segments are aligned along the identity. Therefore, if GARCH is estimated with data that contain different means, the estimated autoregressive parameters pick up the slope of the identity. This effect causes the estimated sum of the autoregressive parameters of the conditional variance in GARCH to grow close to unity as both the sample size and the parameter jump size increases. The effect does not depend on the estimation method. It is not necessary to have a certain underlying stochastic structure that drives the parameter changes; a single deterministic change-point is sufficient.

In simulations of the GARCH model, I show that this effect occurs for sample sizes and jump sizes that are realistic for financial market volatility. Therefore, whenever GARCH is estimated on long-range data sets that contain change-points, it will show an estimated sum of the autoregressive parameters close to one. It will indicate high persistence even though the average persistence within segments of constant parameters may be low.

These results do not imply that the long memory found in financial volatility data is in fact spurious. It may well be that a genuine long-memory process and occasional change-points co-exist in the data. The findings show, however, that in the presence of neglected parameter change-points, GARCH is an inappropriate model for measuring volatility persistence. Before carrying out GARCH estimations on financial volatility data, a thorough change-point study of the data is necessary in order to avoid this "spurious almost-integration" effect.

## Acknowledgements

The main research for this paper was done while the author was visiting the Department of Mathematics at Stanford University. I am deeply indebted to George Papanicolaou. Financial support from the Graduirten Kolleg Komplexe Dynamische Systeme at the University of Bremen, Germany, is acknowledged. Thanks to Ludwig Arnold, Heinz-Otto Peitgen, and Otto Steiger. The paper benefited from discussions with Caio Almeida, Arnold Kim, Mordecai Kurz, Jonathan Mattingly, and Knut Sølna, the participants of the Economic and Social Research Council conference on long memory, structural breaks and stock market volatility, and from the comments of the anonymous referees. Thanks to David Brasington for a native speaker's eye on the text. Remaining errors of any kind are mine.

## References

Andersen, T.G., Bollerslev, T., 1997. Intraday periodicity and volatility persistence in financial markets. Journal of Empirical Finance 4, 115–158.

Andersen, T.G., Bollerslev, T., Diebold, F.X., Ebens, H., 2001. The distribution of stock return volatility. Journal of Financial Economics 61, 43–76.

Andreou, E., Ghysels, E., 2002. Detecting multiple breaks in financial market volatility dynamics. Journal of Applied Econometrics 17, 579–600.

Baillie, R.T., Bollerslev, T., Mikkelsen, H.O., 1996. Fractionally integrated generalized autoregressive conditional heteroskedasticity. Journal of Econometrics 74, 3–30.

Bollerslev, T., 1986. Generalized autoregressive conditional heteroskedasticity. Journal of Econometrics 31, 307–327.

Bollerslev, T., 1987. A conditionally heteroskedastic time series model for speculative prices and rates of return. Review of Economics and Statistics 69, 542–547.

Bollerslev, T., Engle, R.F., 1993. Common persistence in conditional variances. Econometrica 61, 167–186.

Bollerslev, T., Wooldridge, J.M., 1992. Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. Econometric Reviews 11, 143–172.

Bos, C.S., Franses, P.H., Ooms, M., 1999. Long memory and level shifts: re-analyzing inflation rates. Empirical Economics 24, 427–449.

Chernov, M., Gallant, A.R., Ghysels, E., Tauchen, G., 2003. Alternative models of stock-price dynamics. Journal of Econometrics 116, 225–257.

Choi, K., Zivot, E., 2002. Long memory and structural changes in the forward discount: an empirical investigation. mimeo. http://faculty.washington.edu/ezivot/ezresearch.htm.

Diebold, F.X., 1986. Modeling the persistence of conditional variances: a comment. Econometric Reviews 5, 51–56.

Diebold, F.X., Inoue, A., 2001. Long memory and regime switching. Journal of Econometrics 105, 131–159.

Ding, Z., Granger, C.W.J., 1996. Modeling volatility persistence of speculative returns: a new approach. Journal of Econometrics 73, 185–215.

Ding, Z., Granger, C.W.J., Engle, R.F., 1993. A long memory property of stock market returns and a new model. Journal of Empirical Finance 1, 83–106.

Doornik, J.A., Ooms, M., 2000. Multimodality and the GARCH likelihood. Working paper. http://www.feweb.vu.nl/econometriclinks/ooms/papers/

Engle, R.F., 1982. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. Econometrica 50, 987–1007.

Engle, R.F., Bollerslev, T., 1986. Modelling the persistence of conditional variances. Econometric Reviews 5, 1–50.

Engle, R.F., Lee, G.G.J., 1999. A long-run and short-run component model of stock return volatility. In: Engle, R.F., White, H. (Eds.), Cointegration, Causality, and Forecasting: A Festschrift in Honour of Clive W.J. Granger. Oxford University Press, Oxford.

Engle, R.F., Patton, A.J., 2001. What good is a volatility model? Quantitative Finance 1, 237–245.

Fouque, J.P., Papanicolaou, G., Sircar, K.R., Sølna, K., 2003. Short time-scale in S&P 500 volatility. Journal of Computational Finance 6, 1–23.

Francq, C., Roussignol, M., Zakoïan, J.M., 2001. Conditional heteroskedasticity driven by hidden Markov chains. Journal of Time Series Analysis 22, 197–220.

Gallant, A.R., Tauchen, G., 2001. Efficient method of moments, mimeo. http://www.unc.edu/~arg.

Granger, C.W.J., 1980. Long memory relationships and the aggregation of dynamic models. Journal of Econometrics 14, 227–238.

Granger, C.W.J., Hyung, N., 1999. Occasional structural breaks and long memory. UC San Diego Discussion Paper 99–14.

Granger, C.W.J., Teräsvirta, T., 2001. A simple nonlinear time series model with misleading linear properties. Economics Letters 62, 161–165.

Gray, S.F., 1996. Modeling the conditional distribution of interest rates as a regime-switching process. Journal of Financial Economics 42, 27–62.

Hamilton, J.D., Susmel, R., 1994. Autoregressive conditional heteroskedasticity and changes in regime. Journal of Econometrics 64, 307–333.

Kokoszka, P., Leipus, R., 1999. Testing for parameter changes in ARCH models. Lithuanian Mathematical Journal 39, 182–195.

Kokoszka, P., Leipus, R., 2000. Changepoint estimation in ARCH models. Bernoulli 6, 513–539.
Lamoureux, C.G., Lastrapes, W.D., 1990. Persistence in variance structural change and the GARCH model. Journal of Business and Economic Statistics 8, 225–234.
LeBaron, B., 2001. Stochastic volatility as a simple generator of apparent financial power laws and long memory. Quantitative Finance 1, 621–631.
Lobato, I.N., Savin, N.E., 1998. Real and spurious long-memory properties of stock-market data. Journal of Business and Economic Statistics 16, 261–268.
Lumsdaine, R., 1995. Finite-sample properties of the maximum-likelihood estimator in GARCH(1,1) and IGARCH(1,1) models: a Monte Carlo investigation. Journal of Business and Economic Statistics 13, 1–10.
Lumsdaine, R., 1996. Consistency and asymptotic normality of the quasi-maximum likelihood estimator in IGARCH(1,1) and covariance stationary GARCH(1,1) models. Econometrica 64, 575–596.
Mikosch, T., Starica, C., 2004. Nonstationarities in financial time series, the long-range dependence, and the IGARCH effects. The Review of Economics and Statistics 86, 378–390.
Ohannissian, A., Russell, J., Tsay, R., 2003. Using temporal aggregation to distinguish between true and spurious long memory. Working paper. http://gsb-www.uchicago.edu/fac/jeffrey.russell/research/wp.htm.
Press, W.H., Teukolsky, S.A., Vetterling, W.T., Flannery, B.P., 2002. Numerical Recipes in C++. Cambridge University Press, Cambridge.
Sakoulis, G., Zivot, E., 2000. Time-variation and structural change in the forward discount: implications for the forward rate unbiasedness hypothesis. mimeo. http://faculty.washington.edu/ezivot/ezresearch.htm.
Weiss, A.A., 1986. Asymptotic theory for ARCH models: estimation and testing. Econometric Theory 2, 107–131.
Zumbach, G., 2000. The pitfalls in fitting GARCH(1,1) processes. In: Dunis, C. (Ed.), Advances in Quantitative Asset Management. Kluwer, Dordrecht.